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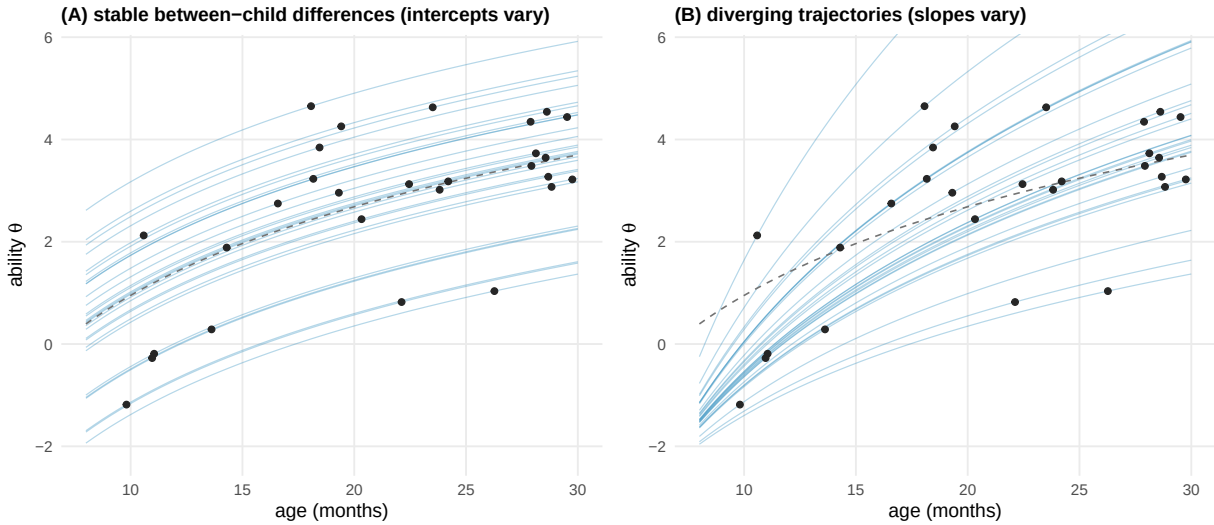


Figure 1: Non-identifiability of growth models in cross-sectional data. Panels show the same simulated data, reflecting the possibility of stable between-child intercept differences (A) vs. varying growth slopes (B).

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## Relation to prior accumulator models [draft for promotion - MCF to review]
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Our model is one member of a family of accumulator models of early vocabulary, and the family is easiest to compare in a single set of coordinates. Writing the latent ability of child i on word j at age t as

$$\eta_{ij}(t) = \log r_i + \log \alpha_i + \kappa_i \log t - \delta_j,$$

every prior model in this tradition is a special case that fixes one or

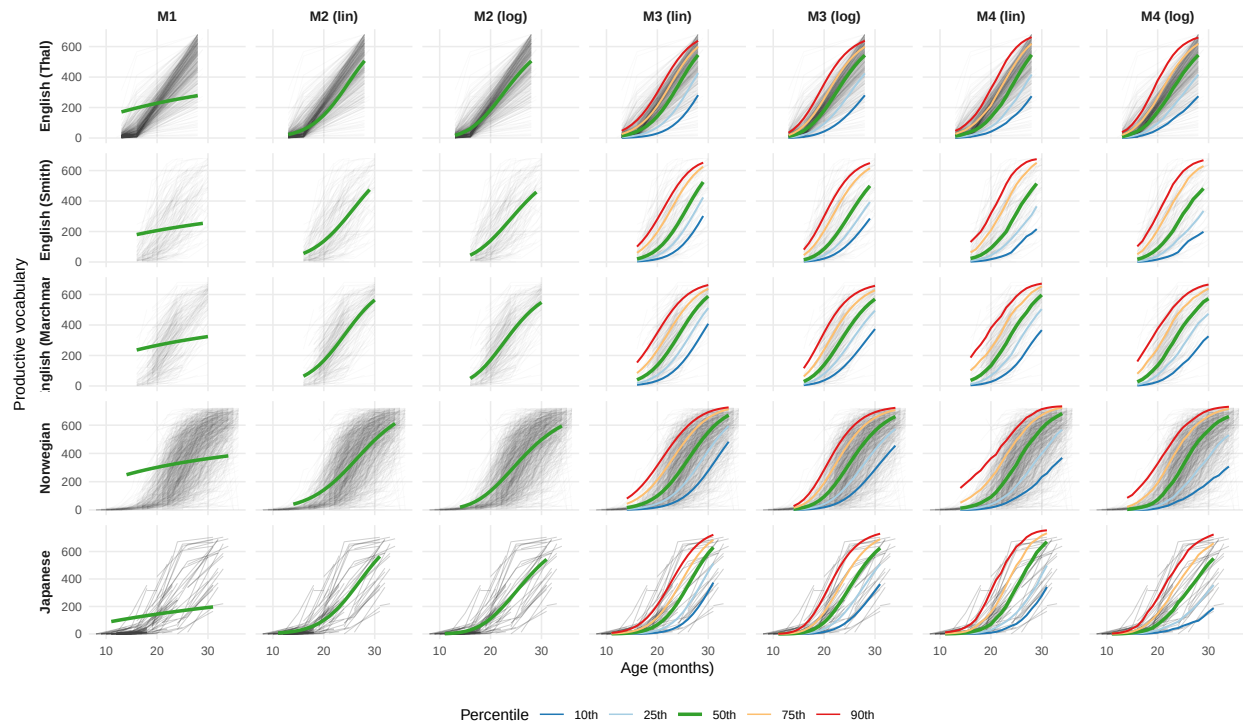


Figure 2

more of three quantities: the between-child variance in efficiency (σ_{α}), the population scaling exponent (κ_{pop}), and - critically - the between-child variance in that exponent (σ_{κ}). Mapping each prior model into these coordinates makes the comparison structural rather than notational.

McMurray [-@mcmurray2007] is a pure accumulator: every word fills at a single unit rate ($\kappa = 1$) and children do not differ ($\sigma_{\alpha} = \sigma_{\kappa} = 0$). Its central insight is that the *vocabulary* curve accelerates even when the underlying rate does not, because ability sweeps through a distribution of word difficulties - the construction reproduced in panel (A) of our schematic. Acceleration here is a population-level property of the difficulty distribution, not a change in the child. This corresponds to our M1.

Mitchell & McMurray [-@mitchell2009] add a leverage term: already-known words speed the acquisition of new ones. They prove that this feedback shifts *when* words are acquired but cannot manufacture acceleration that is not already present in the difficulty distribution. In our coordinates it moves the timing of acquisition without changing κ , and - being non-linear in accumulated vocabulary - it is the one prior model that does not embed cleanly in the linear predictor above.

Hidaka [-@hidaka2013] introduces a rate-change process in which ability grows as t^{D+1} ; his exponent D is exactly $\kappa_{\text{pop}} - 1$ in our notation, so $D > 0$ is population-level acceleration ($\kappa_{\text{pop}} > 1$). This is our M2. The acceleration is shared across children ($\sigma_{\kappa} = 0$). Hidaka relates D to word imageability - i.e. a *per-word* scaling exponent - which is an interesting direction we do not pursue here.

Mollica & Piantadosi [-@mollica2017] model each word as a Gamma waiting-time process in which acquisition occurs after roughly ten "effective learning instances." In our coordinates this is again a pure accumulator ($\kappa = 1$); the per-word variation they capture lives in the Gamma shape parameter, which corresponds to a per-word discrimination (a 2PL extension) rather than to per-child structure.

Kachergis et al. [-@kachergis2022] are the first to free σ_{α} : they cast vocabulary as a Rasch model in which each child has their own latent ability, unifying the accumulator and item-response traditions. This is our M3. But the scaling exponent remains shared across children ($\sigma_{\kappa} = 0$), and the model is proposed as a framework rather than fit to longitudinal individual data.

Reading down this list, the literature progressively frees parameters - McMurray fixes all three, Hidaka frees κ_{pop} , Kachergis frees σ_{α} - but every model holds $\sigma_{\kappa} = 0$.

Model	σ_{α}	κ_{pop}	σ_{κ}	per-word δ_j
McMurray (2007)	0	1	0	free
Mitchell & McMurray (2009)	0	1	0	free + leverage
Hidaka (2013), rate-change	0	$D+1$	0	free
Mollica & Piantadosi (2017)	0	1	0	free
Kachergis et al. (2022)	free	free	0	free
This work (M4)	free	free	free	free

Our contribution is to free σ_{κ} : children differ not only in how efficient they are but in how steeply their learning accelerates. This is the load-bearing addition - it is what lets the model fit the widening fan of trajectories with age (M4 vs M3 in the main text). It is also a falsifiable claim rather than a parameter convenience: if σ_{κ} were near zero the model would collapse onto Kachergis et al., but the estimated per-child slope variance is well above zero and large relative to the efficiency variance σ_{α} (main text, Figure 2 and Results).

Comparing children and language models on one slope axis [draft for promotion - MCF to review]

The comparison to language models in Figure 6 places children and models on a single, literally shared axis: the slope of the acquisition sigmoid in log-experience. Both populations have sigmoidal per-word acquisition curves, and the slope of that sigmoid is directly comparable.

On the child side, our linear predictor makes this slope exact. Differentiating $\eta_{ij}(t)$ with respect to $\log t$ gives κ_i : the logit-probability that a child produces a word rises linearly in log-age with slope equal to that child's scaling exponent. The per-child sigmoid slope *is* κ_i , by construction.

On the model side, @chang2022 fit a four-parameter logistic to each (model, word) pair, describing how the model's surprisal for a word falls over training. Re-expressed as the fraction of a word's surprisal reduction achieved so far, their fit is a sigmoid in $\log(\text{training steps})$ with slope $1/\text{scale}_w$. Dividing by $\ln 10$ converts this to natural-log units - the same units as κ_i - so the two slopes can be read off the same axis.

The result is a striking separation (Figure 6). Language models cluster near a slope of one - the unit-accumulator boundary, where each log-unit of additional experience yields a fixed increment of learning. Children sit in a broad distribution roughly an order of magnitude higher. On the shared axis, children's word learning accelerates with experience in a way that these models' does not.

Three caveats bound this comparison. First, the variability is asymmetric: children's slopes vary across *children*, whereas a single model is a deterministic learner, so model slopes vary across *words* within a run rather than across replicate learners. Second, the steep child slope is the *per-child* quantity; a researcher fitting a single sigmoid to cross-sectional Wordbank data per word would recover a shallower slope, because marginalizing over the population attenuates it. Third, the models in @chang2022 were trained on adult written text rather than child-directed speech, so part of the gap could reflect the input distribution rather than the learner. Models trained directly on CHILDES (Feng et al., forthcoming) provide the matched-input test: if the gap largely persists, it is structural; if it shrinks substantially, the input distribution is doing more of the work than we currently attribute to it.

<!-- [BIB NEEDED] Feng et al. (CHILDES-trained GPT) - no verified entry yet; named in text only. Daunizeau (2017) logistic-Gaussian approximation is used in the derivation note but not needed for this prose. -->

Connection to neural scaling laws [draft for promotion - MCF to review; most speculative of

The per-word picture above connects to a more familiar aggregate phenomenon: neural scaling laws. A model's loss falls as a smooth power law in the amount of training data, and this descent is remarkably regular - different runs, and different model sizes, trace nearly the same curve (Kaplan et al., 2020; Hoffmann et al., 2022). Our per-word analysis is the micro-level counterpart of that regularity. Each word is acquired along a sigmoid whose slope sits near the unit-accumulator value, and the spread of those slopes - across words and across runs - is small. The aggregate smoothness of scaling laws and the per-word $\kappa \approx 1$ we recover are two views of the same property: language models convert experience into knowledge as near-pure accumulators, with little instance-to-instance variation.

Children depart from this picture on both axes at once. Their per-instance scaling is far steeper ($\kappa \approx 10$), and they vary substantially from one another in that exponent (σ_κ well above zero). The "data gap" between children and language models [Frank2023; Warstadt2023] is therefore not only that children reach adult competence from orders of magnitude less input. It is that children convert experience into knowledge through a different scaling relationship - steeper, and far more variable across individuals - than the one that scaling laws describe for models.

This connection is analogical rather than formal: scaling laws characterize aggregate loss, whereas our κ characterizes per-item acquisition, and the two need not move together. We offer it as a framing for why the data gap may be harder to close than raw data counts suggest.

<!-- [BIB NEEDED] Kaplan et al. (2020) and Hoffmann et al. (2022) are named in text but have NO verified bib entries yet - do not promote with live @ markers until added. -->

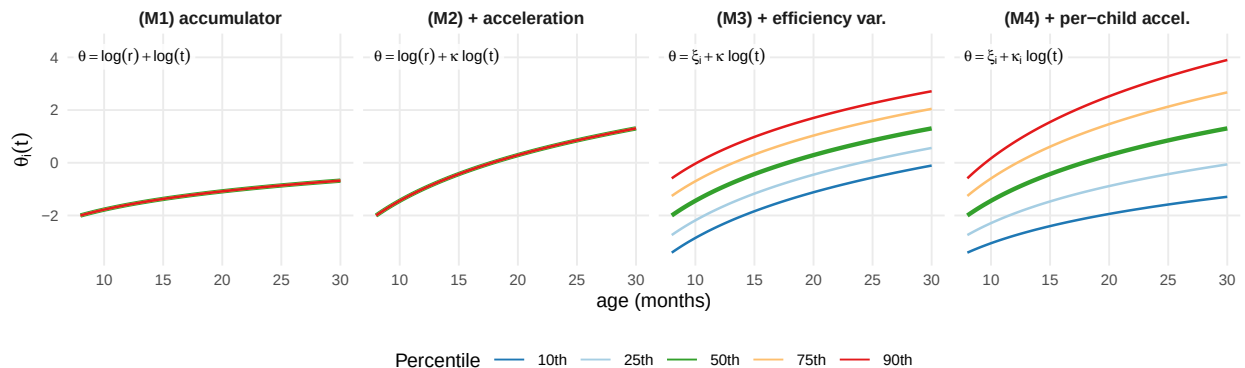


Figure 3

Table 1

Model	Age	Δ BIC				
		Thal	Smith	Marchman	Norwegian	Japanese
M1. accumulator	—	703,172	273,687	251,166	2,281,553	94,461
M2. + acceleration	linear	206,517	179,838	134,563	1,377,575	41,138
M2. + acceleration	log	204,895	178,437	133,843	1,347,973	39,023
M3. + efficiency var.	linear	35,832	17,396	13,729	130,599	3,829
M3. + efficiency var.	log	34,195	15,333	13,113	115,089	3,206
M4. + acceleration var.	linear	1,134	553	0	4,301	169
M4. + acceleration var.	log	0	0	30	0	0

Table 2

Dataset	N	Efficiency (ξ)	Acceleration (ζ)
English (Thal)	651	0.96	>0.99
English (Smith)	312	0.30	0.99
English (Marchman)	313	0.99	0.99
Norwegian	1667	>0.99	0.86
Japanese	176	0.85	0.99
Pooled	3119	0.98	0.96

Table 3

Source	Sample	Channel	n	tok/hr	μ SD	tok/mo	σ_r
Hart & Risley	all kids pooled	mother CDS	42	1,148	546–2,410	419K	0.74
Hart & Risley	within-stratum pooled	mother CDS	42	1,148	631–2,086	419K	0.60
Soderstrom & Wittebolle	all kids pooled	all speech	12	967	622–1,502	353K	0.44
Sperry	all kids pooled	adult-to-child	42	1,537	901–2,622	561K	0.53
Sperry	within-stratum pooled	adult-to-child	42	1,537	974–2,425	561K	0.46
Weisleder & Fernald	all kids pooled	mother CDS	29	458	197–1,065	167K	0.84
Bergelson 2018 (SEEDLingS, LENA)	US daylong, 6–7 mo	all adult (AWC)	44	1,101	767–1,580	402K	0.36
Casillas 2020 (Tseltal Mayan)	Chiapas daylong	child-directed	10	720	—	263K	—
Coffey et al. 2024 (plausible envelope)	cross-cultural bounds	CDS to all-input	—	—	4–7,232	—	—

Estimating the input-rate prior: units and conversions [draft for promotion - MCF to review]

The model represents each child's input as a log rate,
 $\log r_i \sim \mathcal{N}(\mu_r, \sigma_r^2)$, measured in **adult tokens**
per waking hour. @tbl-input-rates collects published estimates of this
rate, expressed in shared units so they can be compared and so that the
implied between-child spread σ_r is visible on one scale.

Conventions used throughout:

- **Tokens/hour to tokens/month.** We multiply by $H = 365$ waking hours
per month ($\text{hr/day} \times 30.44 \text{ days/mo}$), the model's

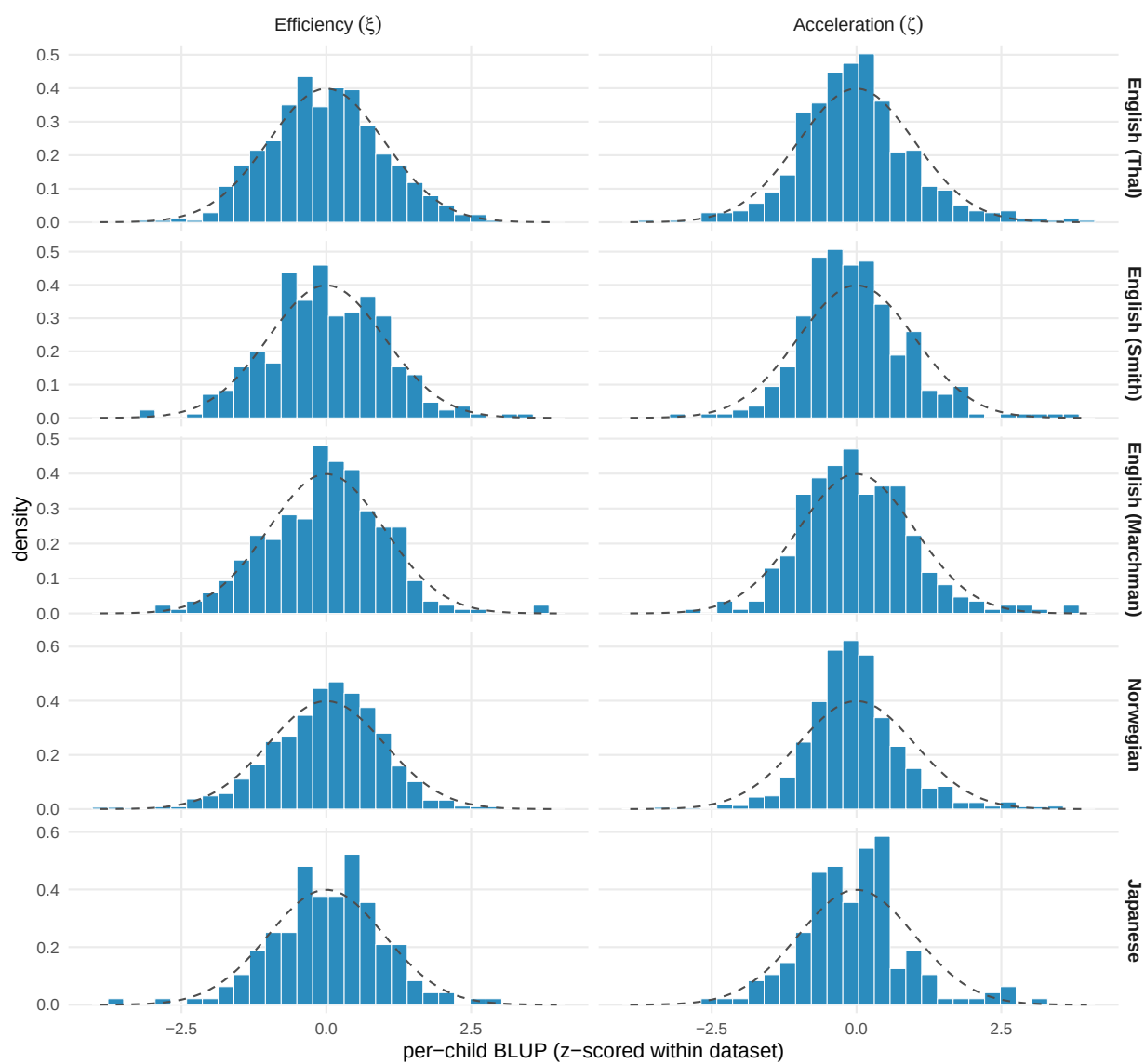


Figure 4

constant.

- **σ_r is on the log scale** - the SD of $\log(\text{tokens/hr})$ across children within a sample. Its exponential e^{σ_r} is the multiplicative factor for a one-SD shift in input (e.g. $\sigma_r = 0.53 \rightarrow 1.7\times$), and the 16th-84th percentile spread is $e^{2\sigma_r}$.
- **Channel.** Studies count different speaker sets; we take the channel closest to "tokens reaching the child" - adult-to-child for Sperry, all-adult LENA word counts for SEEDLingS, and maternal CDS for Hart & Risley and Weisleder & Fernald.
- **Pooled vs. within-stratum.** For Hart & Risley and Sperry we report σ_r both across all children (which mixes in SES and site gradients) and within stratum (subtracting each band's mean first); the latter is the cleaner within-population estimate.
- **Minutes to tokens.** Where a source reports minutes of speech per hour rather than token counts (Casillas; the Coffey bounds), we convert at ≈ 200 words/min (0.3 s/word, the Brent-corpus CDS rate used by @coffey2024); those rows are accurate to roughly $\pm 20\%$.

Rows marked *computed* are recomputed here from per-dyad data; *literature* rows are taken from the curated validation set in `input_estimation/`.

`## Analytic input-rate ( $\sigma_r$ ) sensitivity [draft for promotion - MCF to review]`

The input/efficiency variance split is identified only by the external pin on the population SD of log input rate, σ_r - we do not observe input in the longitudinal data. Because every reported input share therefore rests on a quantity we assume rather than estimate, we map how the conclusion moves as σ_r ranges over its plausible values (panel B of @fig-io-partition).

The data identify the *total* between-child intercept variance, $\sigma_{\xi}^2 = \sigma_{\alpha}^2 + \sigma_r^2$ (the spread of children's ability at the reference age a_0), very precisely; the pin only decides how that fixed total is *split* into an efficiency part σ_{α}^2 and an input part σ_r^2 . Holding σ_{ξ}^2 at its fitted value, the input share is then a closed form,

$$1 - \pi_{\alpha}(\sigma_r) = \frac{\sigma_r^2}{\sigma_{\xi}^2},$$

which we evaluate across the plausible σ_r range *without refitting* - the analytic sensitivity curve.

This is first-order: it treats σ_{ξ}^2 as itself invariant to the pin. Refits at off-pin values reveal a small second-order feedback - pinning a larger σ_r slightly *shrinks* the estimated total variance (for English,

σ_{ξ}^2 falls from 3.57 at $\sigma_r=0.53$ to 3.22 at $\sigma_r=0.80$). The analytic curve therefore slightly *understates* the input share at large σ_r (18% analytic vs. 20% by refit at $\sigma_r=0.80$), an error of about two points of π_{α} in the conservative direction for the size of the input effect. The open points in panel B are these refit anchors; the gap between point and curve is the second-order term.

We use the analytic curve, anchored by the few refits we have, rather than a dense refit sweep: the qualitative conclusion - input accounts for a modest share even at the top of the plausible σ_r range - is unaffected by the ~2-point correction, whereas a full sweep of the Norwegian model alone would cost roughly half a day of compute per pin value.